



WRIST CONTROL for Vallox

Fresh air at home, right on your wrist.

A Wear OS app for Vallox heat recovery ventilation. See how fresh the air is, switch modes in two taps, and let the app tune the unit to the way your home breathes.



See your air

CO₂, humidity and temperatures at a glance, coloured green, yellow or red.

Smart modes

Every mode you pick on the watch sets its own fan speed and CO₂ limit, and Optimize tunes Automatic to your home.

Know the cost

A weekly summary of air quality and fan electricity, and why it changed.

WHY IT PAYS OFF

The same fresh air, with less fan.

	Vallox unit on its own	With Wrist Control
Modes	Away and Automatic share one fan speed, so one of them is always a compromise.	Each mode sets its own fan speed and CO2 limit the moment you switch.
Tuning	Set by hand once, the same for every home and every season.	Optimize fits Automatic to your home's last 7 days, shows a forecast and can restore your old values.
Your week	Current values and charts to read yourself.	Fresh-air share, fan electricity and why it changed, in one glance.

One home, one week, six ways to run the unit

CO2 shows how stale the air is: fresh up to 800 ppm ●, stuffy to 1200 ●, poor above ●. Fan power grows roughly with the cube of speed, so 70 % uses about 60 % more electricity than 60 %.

	Fan electricity 60 % all day = 100	Morning CO2 typical, ppm	Busy hours CO2, ppm
Fan at 70 % all day	159	649	748
Fan at 60 % all day how this home ran	100	687	803
Optimize: Clean air	133	649	719
Optimize: Balance	76	691	765
Optimize: Quiet	19	836	915
Automatic, basic settings Away 30 %, limit 800	18	862	941

52 % less

fan electricity than running at 70 % all day, with about the same air. That is Optimize Balance.

Fresher busy hours

than 60 % all day: 765 instead of 803 ppm, while using 24 % less electricity.

81 % less

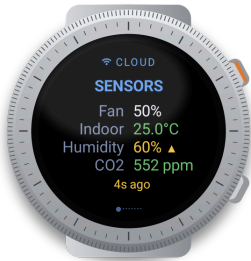
fan electricity with Quiet, and a quieter home. Mornings rise to about 836 ppm.

Estimates from one home's 7-day history, made with the same model Optimize uses. Fan electricity only, relative to running at 60 % all day. Heating energy is not included: less air also means less heat to recover in winter. "Busy hours" are the stuffiest 10 % of the time.

THE APP

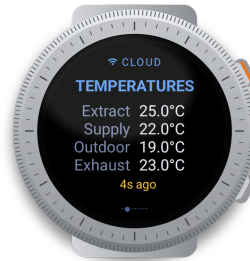
Everything about your air, one swipe apart.

Swipe left and right between pages. Tap a page to act on it.



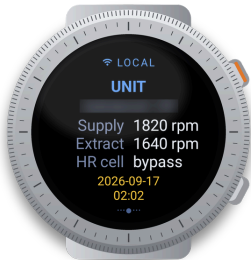
Sensors

How the air in your home is right now: fan speed, temperature, humidity and CO2, coloured by comfort. Tap to see the comfort scales.



Temperatures

The four air streams: extract air from the rooms, supply air to them, outdoor air coming in and exhaust air going out.



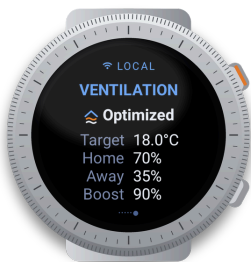
Unit

Your unit's name, the speed of both fans, whether the heat exchanger is recovering heat or bypassed to cool, and the unit's own clock.



Filters

When the filters were changed and when the next change is due. After fitting new ones, tap Changed twice: the date is saved in the unit.



Ventilation

The mode the unit runs now and the fan speed of each mode. Tap to switch modes.



Mode

Pick a mode here, not on the unit's panel: the watch also sets the values that belong to it. The running mode spins.



Settings

Where the numbers come from, how often they refresh, which alerts are on, the temperature range and which pages are shown.



On your watch face

Indoor, outdoor or supply air temperature, CO2, humidity, fan speed or days to the filter change on your watch face, updated without opening the app.

Home and away

At home the watch talks to the unit over Wi-Fi. Away, it uses the Vallox cloud and your MyVallox account.

Alerts that don't nag

One notice when indoor air leaves the comfort range, with the same scale as in the app. One when filters are due. No repeats.

Your pages

Hold a page to reorder or hide pages. Keep only what you look at. Ventilation and Settings always stay.

Sign in from the phone

No typing on the watch: the phone app sends your MyVallox login to the watch. English, Suomi, Lietuvių, Polski.

SMART MODES · THE HEART OF THE APP

Switch modes on the watch, and every mode runs the way it should.

Your Vallox keeps **one fan speed for both Away and Automatic**, and one CO2 limit (the point where it starts adding air) for every mode. The unit's panel switches only the mode, so the unit keeps whatever values were set last. The watch switches the mode **and** sets that mode's own values, in the same tap.

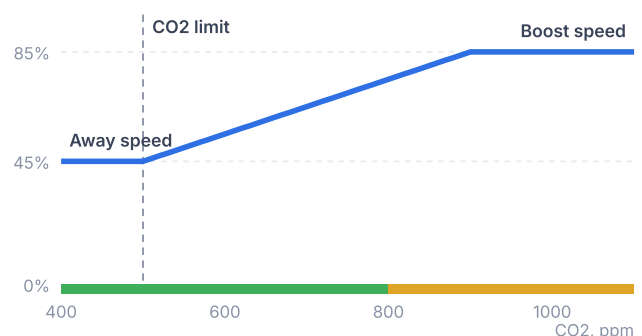
SAME TAP	ON THE UNIT'S PANEL	ON THE WATCH
You leave home tap Away	Away keeps the speed tuned for Automatic, 40 %: about 2.4 times the fan electricity for an empty house.	Away runs at 30 %, the right speed for an empty house.
You come back tap Automatic / Optimized	Automatic starts from the Away speed of 30 %: mornings reach about 862 ppm , stuffy.	Optimized returns to your Optimize settings: mornings about 691 ppm.

Mode	Use it when	The watch also sets	Lasts
At home	People are home and you want a steady fan	CO2 limit 800 ppm	Until you switch
Away	The house is empty	Fan 30 %, CO2 limit 800 ppm	Until you switch
Optimized first in the list	You want the fan to follow the air	Your Optimize settings (fan at rest 45 %, limit 600 ppm before you optimize). Slows to 30 % when nobody is home, to 35 % when winter air is dry; offers a Boost when humidity jumps	Until you switch
Boost	Cooking, sauna, guests	Nothing	60 min, then back to the previous mode
Custom	Fireplace or special use	Nothing	Its timer, then back
Auto last in the list	You want the Vallox factory behaviour	Factory values: fan 30 %, CO2 limit 800 ppm	Until you switch

How Automatic decides

Two settings decide it. The **Away speed** is the lowest speed the fan runs at. The **CO2 limit** is the point where the unit starts adding air: above it, about 1 % of fan speed for every 10 ppm, up to the Boost speed.

A lower limit or a higher Away speed means fresher air and more fan. Finding the right pair for your home is what Optimize does.



Example: Away speed 45 %, CO2 limit 500 ppm, Boost 85 %.

OPTIMIZED · AUTOMATION

Optimized adjusts itself while you live.

While **Optimized** is on, the app checks four rules after every reading. It never drives the fans directly: two rules only change the **fan at rest** (the Away speed Automatic starts from), and the unit's own CO2 control keeps working between readings. The third rule only offers a Boost, the fourth opens or closes the bypass. The numbers were checked against a year of real history (2025-09-15 - 2026-09-14, 47,789 records every 10 min).

Rule	When	What happens	Ends when	In a year of history
Nobody home	CO2 at most 460 ppm for 90 min without a break, no humidity jumps	Fan at rest 45 → 30 %	CO2 rises 80 ppm above its lowest value, or humidity jumps	89 times, about 1.7 a week, median 1.7 h
Dry air	Below +5 °C outdoors, indoor humidity under 38 % in two readings in a row, CO2 at least 100 ppm under the limit	Fan at rest → 35 %	Humidity rises above 43 %	6 times in a winter, about 67 h in total
Boost offer	Humidity up 8 % within about 30 min	Notification "Cooking or shower?" - tap to Boost	At most once an hour	About 220 a year, 4 a week
Temperature	Indoors outside the range in Settings	Closes the bypass below it, opens it above	Each state at least 1 h	New, 2026-10

Why the fan at rest, not the mode

"Nobody home" does not switch to Away. It only lowers the speed Automatic starts from. When people come back, nobody has to wait for the watch: the unit sees CO2 rising and adds air by itself, and at the next reading the app restores the Optimized value. The same holds for "Dry air".

Boost and Custom

While a temporary mode runs, the automation changes nothing and waits until the unit is back in Optimized.

Week schedule

If the unit's week schedule is on, it would switch the mode at set times and quietly end Optimized. The first time you pick Optimized, the app asks once: pause the schedule while Optimized is on? With Pause the schedule is switched off - its events are kept - and switched back on when you choose another lasting mode. The app restores only what it paused itself. Under Optimized a line says "Schedule paused" or "Schedule on".

OPTIMIZED · HOW OFTEN, AND WHAT NEVER CHANGES

The watch looks when something happens.

Data refresh: Auto	Reading
Hours when your home is usually busy	every 20 min
Ordinary hours	every 30 min
Night, 1-5 h	every 60 min
A change was just seen (humidity +3 % or CO2 +100 ppm since the last reading)	every 15 min for an hour

The busy hours are learned for each unit from its own history (when Optimize or Statistics downloads it) and from the changes the watch sees. Until there is enough data: 6-8 and 19-22 h. The default is a fixed 1 h; Auto, 20 min, 30 min or Only in app can be chosen in Settings → Data refresh. Auto drains the battery faster. With Only in app nothing runs in the background: no notifications and no automation. At night Android may group the readings.

What the app changes

On its own only these: the **mode**, the **Away speed**, the **CO2 limit**, the **week schedule switch**, and while Optimized runs, the **bypass lock** and the **supply air target** of Automatic. When you ask, it also writes the filter change date and corrects the unit's clock.

What it never changes

The Home and Boost speeds, humidity control, the heater and Custom settings. Optimize stays within a safe range (fan at rest 20-70 %, CO2 limit 500-1000 ppm), and Restore brings back the previous values. Leaving Optimized hands the bypass and the supply air target back as they were.

Without a connection

The app changes nothing, and the unit keeps running with the last settings.

Temperature range

Settings → Temperature sets the range, by default 22.5-23 °C. Below it the bypass is closed and the heat exchanger keeps the heat; above it, with outdoors at least 2 °C cooler, outdoor air cools the home. The electric heater is not used. Each state lasts at least an hour, so small swings do not move the damper. Let Vallox decide in the Optimize window turns it off.

Planned

Per-unit thresholds: the "empty home" CO2 level and the humidity jump worked out for each unit from its own history - today they are one apartment's numbers. **Summer cooling with the fan:** more air when the night is cooler outside than inside, less on a hot day. **Weather forecast** - once there is summer experience.

The full paper

The same logic with the Vallox control curve, the mode recipes and Optimize step by step - for installers:

[How the automation works \(PDF, English\)](#)

[Kaip veikia automatika \(PDF, lietuviškai\)](#)

OPTIMIZE

Settings made for your home, not for an average one.

Every home produces CO2 differently: how many people, how big the rooms, when you sleep and cook. Optimize reads your unit's last 14 days, learns how much fresh air your home needs at each time of day, and picks the Away speed and CO2 limit for Automatic.



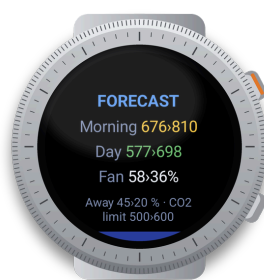
1 · Analyse

Tap Analyse. The watch reads the unit's 14-day history: from the unit at home, from the Vallox cloud when away.



2 · Choose a goal

Clean air (about 600 ppm), Balance (about 700) or Quiet: less air, less fan.



3 · See the forecast

Morning and daytime CO2 and fan speed, now > after. Nothing changes until you tap Apply.



4 · Check after a day

The watch compares the forecast with the real morning. Restore brings back your old values.

HOW IT WORKS

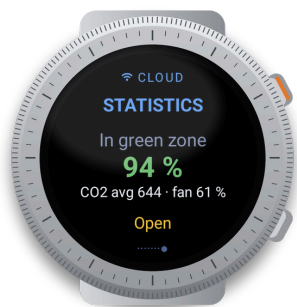
- **Learns your home.** For the morning (6-10), day (10-18), evening (18-24) and night, it measures how fast your home adds CO2, from CO2 and fan speed in the history. It takes the middle of the daily values, so one unusual day - guests, windows shut - does not move the plan.
- **Tries every option.** It simulates each Away speed from 20 to 70 % and CO2 limit from 500 to 1000 ppm and predicts where CO2 settles.
- **Picks the lowest total cost.** Fan electricity (it grows with the cube of speed), heat carried out with the air in the heating season (the heat exchanger returns most of it; how much is measured from your unit's temperatures) and a penalty for every ppm above the goal. So a plan may let CO2 rise a little above the goal at peaks when that saves a lot of energy.
- **Shows the cost.** The forecast has an Energy line against your current settings, and warns when a plan would use more than 1.5 times the energy: a lighter goal then uses less.
- **Goals,** typical / busy hours: Clean 620 / 720 ppm · Balance 700 / 800 · Quiet 850 / 1000.
- **Stays safe.** It never writes values outside those ranges, and keeps your previous values for Restore.
- **After Apply** the window starts with Optimized and the goal, the values it wrote, plan and actual, then Restore and New plan. Below it is the temperature range of the Optimized automation.

Example home: Balance chose Away speed 40 % and CO2 limit 500 ppm.

STATISTICS

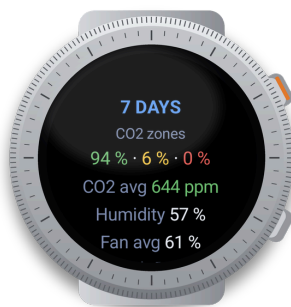
Your week: was the air good, and what did it cost?

Statistics reads the unit's history and answers three questions: how fresh the air was, how much the fans used compared with last week, and why that changed.



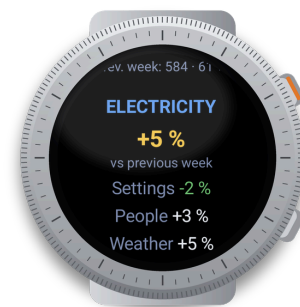
Fresh air share

How much of the week CO2 stayed up to 800 ppm, with average CO2 and fan speed.



The week in detail

Time in fresh, stuffy and poor air, and average CO2, humidity and fan next to last week.



Electricity and why

Fan electricity against last week, split into three reasons.

Settings

You changed a mode's speed or ran a different mode.

People

The unit added air because CO2 rose: more people at home, guests.

Weather and other

Humidity boosts, summer cooling, manual boosts.

HOW IT WORKS

- **Electricity from fan speed.** Fan power grows roughly with the cube of speed, so the change is shown in percent, not kWh.
- **Reads the settings from the history.** For every hour it finds the mode's base speed and the CO2 limit that was in use, so changes made on the panel or in another app are counted correctly.
- **Splits the change.** Base speed counts as Settings, the extra added for CO2 as People, everything else as Weather and other.
- **Heating season.** When it is cold outside, it also shows heat loss per degree of cold, which takes the weather out of the comparison.

Requirements

Wear OS 4 or newer. A Vallox unit with MyVallox: cloud account or access on your home network.

Privacy

No ads, no analytics, no tracking. Your account stays on the watch, password encrypted, sent only to the Vallox cloud.

Price

Free to download. Pro (smart modes, Optimize, Statistics) is a one-time purchase with 14 days free to try.

Independent

Wrist Control for Vallox is not made, endorsed or supported by Vallox Oy. Vallox and MyVallox are trademarks of Vallox Oy.

Questions, feedback, a photo of something wrong
hello@tinyowllabs.com

App and privacy policy
tinyowllabs.com/wristcontrol

Independent developer
[TinyOwl](#)

STATISTICS · LAST YEAR

Better or worse than a year ago, with the weather taken out.

The Vallox cloud keeps a year of your unit's history. Statistics downloads it once, keeps a short summary of every week on the watch, and compares your week with the same week last year.

On the Statistics page

"Energy vs last year": the change in percent, and the fresh-air share now and a year ago, for example "fresh air 76 % vs 74 %".

In the window

The LAST YEAR section: the same number, the total change and the weather part, and last year's CO₂, fan speed and outdoor temperature.

Needs the cloud

The year of history comes only from the Vallox cloud. Without a cloud account the section says so, and the week-to-week comparison still works.

HOW IT WORKS

- **The same week.** The week that ended on the same weekday 52 weeks ago, within a day. If it is missing, the week 51 weeks back.
- **Weather taken out.** The change in fan electricity is split as on the previous page; the weather part is removed, so the number shows what your settings and your home changed.
- **Kept on the watch.** The cloud keeps exactly 365 days. The weekly summaries stay on the watch, so the comparison keeps working when old days drop out of the cloud.

APPENDIX

Getting around



Swipe sideways
next page



Tap a page
open its window



Swipe sideways in
a window
back



Swipe up
close the app



Turn the bezel
scroll or flip pages

1

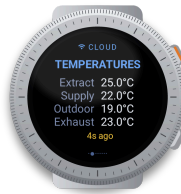


tap
→



Comfort
scales

2

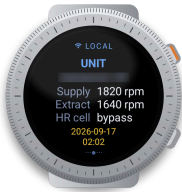


tap
→



Mode

3



tap
→



Refresh

4



tap
→



Filters
changed

5

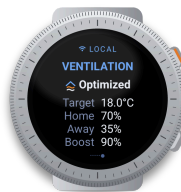


tap
→



Settings

6



tap
→



Mode

7

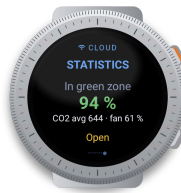


tap
→



Optimize

8



tap
→



Statistics



Hold any page

Pages turn into round cards. Drag to change the order, tap to finish. Hide pages in Settings.

APPENDIX

Settings

Tap the SETTINGS page. The window scrolls; turn the bezel or swipe up.



Cloud · My account

Sign in here or from the phone app. Under the button: the account, and a warning if Vallox rejects it

Unit · Connection

Which unit, and the path: Auto, home Wi-Fi only or cloud only. Without the cloud, Find units searches your network

Data refresh

1 h (default), 20 or 30 min, only in the app, or Auto: every 20-60 min, every 15 after a change

Unit clock

The unit keeps its own time. Ask to fix, fix automatically, or don't check

Alerts

Notifications for temperature, humidity, CO2 and filters

Pages

Show or hide each page. Ventilation and Settings always stay

How it works · Language

Short help on every feature. English, Suomi, Lietuvių, Polski

Send test alerts

Five sample notifications, to see how they look on your watch

Questions, feedback, a photo of something wrong
hello@tinyowllabs.com

App and privacy policy
tinyowllabs.com/wristcontrol

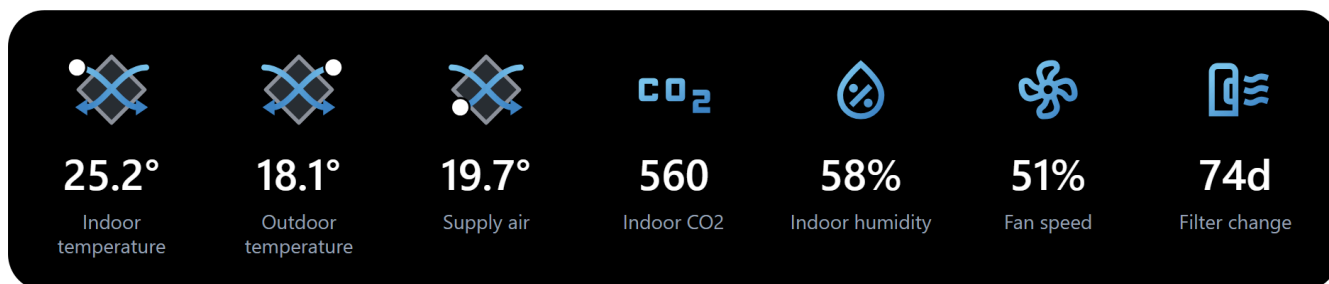
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APPENDIX

On your wrist without opening the app

Complications

Touch and hold the watch face, tap Customize, pick a slot and choose Wrist Control. Tap a complication to open the app.



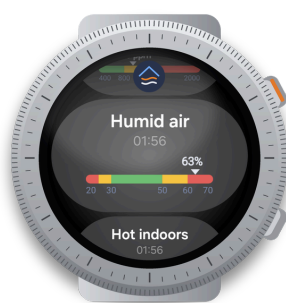
The diamond is the unit's heat exchanger, drawn like on the unit itself. The white dot shows where the temperature is measured. Filter change shows days left, with a minus when overdue.

Notifications



Temperature

below 18 or above 27 °C



Humidity

below 25 or above 60 %



CO2

from 800 ppm, again from 1200



Unit clock

more than 5 min off; tap to fix

One notice when a value leaves its range, no repeats; it disappears when the value is back. The scale is the same as in the app, so you see how far out you are. Filters get a notice when a change is due. Temperature, humidity, CO2 and filter notices can be turned off in Settings → Alerts. The clock notice cannot: a wrong unit clock spoils the history that Optimize and Statistics rely on. Choose Don't check to silence it. A notice that suggests a mode - high CO2 or humidity, or "Cooking or shower?" with Tap to Boost - opens the Ventilation page, so you see the mode change.